

# ReBaseGuard

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*Academic research brief - repository presentation artifact, not a peer-reviewed publication or accepted manuscript.*

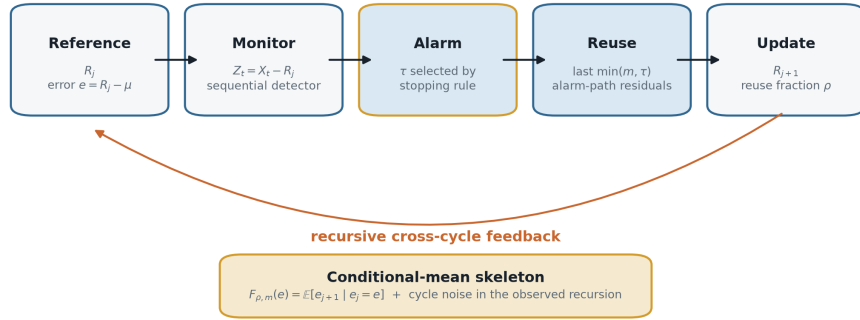
## 1. Problem and motivation

Sequential drift monitors are commonly described one alarm at a time, but real systems may repeat the cycle: monitor, alarm, update the reference, and monitor again. A subtle problem appears when the update reuses observations that participated in the alarm. Those observations were selected through a data-dependent stopping event, so they need not behave like fresh reference data. Their reuse changes the next reference, which changes the next monitoring cycle, creating recursive selection feedback.

ReBaseGuard isolates this **stopping-selected recursive re-baselining** mechanism. The project asks how the feedback changes local reference dynamics, when it becomes unstable in a deterministic conditional-mean description, and whether a stability-aware reuse rule improves monitored behavior in scoped experiments.

### Stopping-selected recursive re-baselining

The next reference reuses observations selected by the alarm stopping time



Conceptual diagram only. It defines the mechanism; it is not proof or empirical evidence.

Figure 1. Alarm-participating observations update the next reference and recursively affect later cycles.

## 2. Core mechanism

Let  $e$  denote the current reference error and let  $F(e)$  be the deterministic conditional mean of the next reference error. For a reuse fraction  $\rho$ , the one-observation stopped-selection derivative has the form

$$F'_{\rho}(0) = \rho (1 - \Gamma), \quad \Gamma = E_0[Z_{\tau} T_{\tau}].$$

The gain  $\Gamma$  measures how the terminal reused residual covaries with the stopped path score. If  $\Gamma > 2$ , full reuse makes the zero fixed point locally repelling in the deterministic mean map. This is a local mathematical statement, not a claim that the noisy monitoring process is globally or operationally unstable.

### 3. Main theoretical results

| Result                            | Evidence                               | Scoped conclusion                                                    |
|-----------------------------------|----------------------------------------|----------------------------------------------------------------------|
| Gaussian CUSUM derivative         | Human theorem + Lean-checked spine     | Stopped-selection derivative identity for the frozen detector        |
| $\text{Gamma\_CUSUM} > 2$         | Outward-rounded Arb certificate        | Local repulsion at full reuse for the deterministic mean map         |
| Symmetric two-chart SR derivative | Human theorem + conditional Lean spine | Same stopped-score structure for the authoritative SR model          |
| $\text{Gamma\_SR} > 2$            | Post-closure Arb certificate           | $\text{Gamma\_SR}$ lies in $[5.800391799508442, 28.781285803081492]$ |
| Period-two skeleton               | Rigorous deterministic certificate     | Locally attracting period-two orbit of the conditional-mean skeleton |
| Finite-window extension           | Human theorem + deterministic analysis | Protocol-specific m-rho local-stability boundary                     |

### Stopped-selection derivative and local instability

The formal derivative and rigorous numerical bound are separate evidence streams

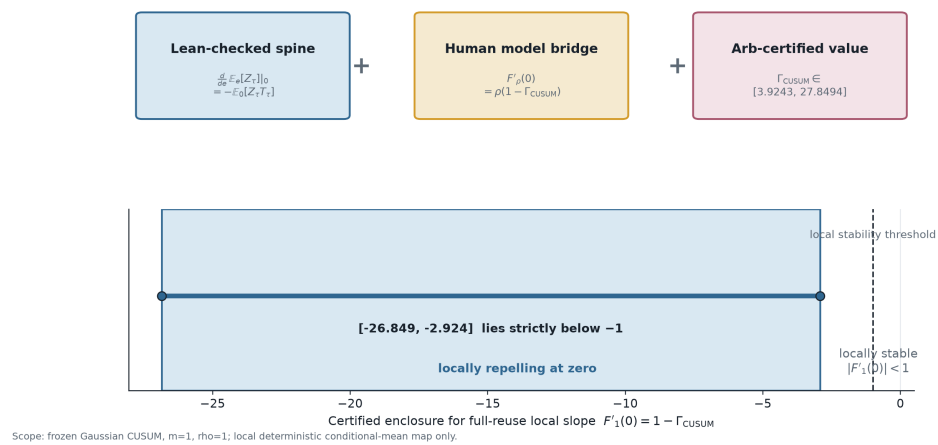


Figure 2. The derivative theorem, human model bridge, and Arb enclosure play distinct roles in the CUSUM instability result.

The Lean formalization checks the encoded differentiation spine and its algebraic consequences. Human proofs carry stopped change-of-measure, measurability, tail, integrability, and domination obligations not wholly discharged in Lean. Arb interval arithmetic independently certifies numerical enclosures; it does not prove differentiation under the expectation. The SR certificate is specific to the authoritative symmetric two-chart detector and does not imply a detector-general or distribution-general theorem.

For a reuse window of size  $m$ , the authoritative random-window convention includes an exact short-cycle correction. Its derivative yields a critical reuse boundary  $\rho_{\text{c}}(m)$ . The full-reuse crossing is bracketed at  $m$  in  $[70, 72]$ , but this boundary belongs to a local deterministic map rather than an operational phase-transition theorem.

## 4. Stability-aware reuse policy

The frozen P3 policy converts the local-stability analysis into a conservative reuse allowance:

$$\text{rho\_P3}(m) = \min(1, 0.8 * \text{rho\_c, L95}(m)).$$

It uses 80% of the simultaneous lower-95% boundary and clips the result at one. At  $m = 1, 20, 70, 100$ , its reuse fractions are 0.053642, 0.245418, 0.781994, and 1.0.

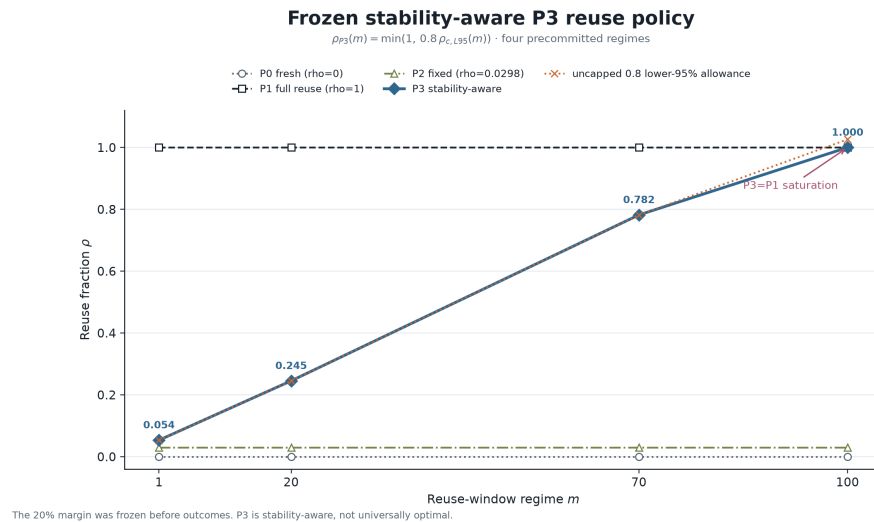


Figure 3. P3 follows an uncertainty-aware stability allowance across the four frozen regimes.

Under the frozen Gaussian CUSUM policy protocol, P3 improved reference-state mean-squared error and false-alert burden against full reuse in active regimes, while passing the primary non-inferiority family. The result is deliberately scoped: P3 equals full reuse at the saturated  $m=100$  regime, P2 retains descriptive advantages at  $m=70$  and  $m=100$ , and two secondary stricter conditions fail. These findings do not establish universal safety or optimality.

## 5. Semi-real validation

Public sequential streams were evaluated task by task without pooling samples. The retained campaign record is Stage E 0/3, V2 1/3, and V3 2/2, giving three supporting tasks against two required by the internal protocol. Unsuccessful tasks remain visible. This is scoped semi-real evidence, not production deployment validation, and policy behavior remains regime-dependent.

## 6. Negative result on the operational crossing

A pre-specified study asked whether the deterministic full-reuse crossing near  $m=71$  produced a corresponding operational transition. It did not: 0/4 preselected metrics peaked at the crossing and 4/4 were monotone in  $\log m$  under the frozen Gaussian CUSUM protocol.

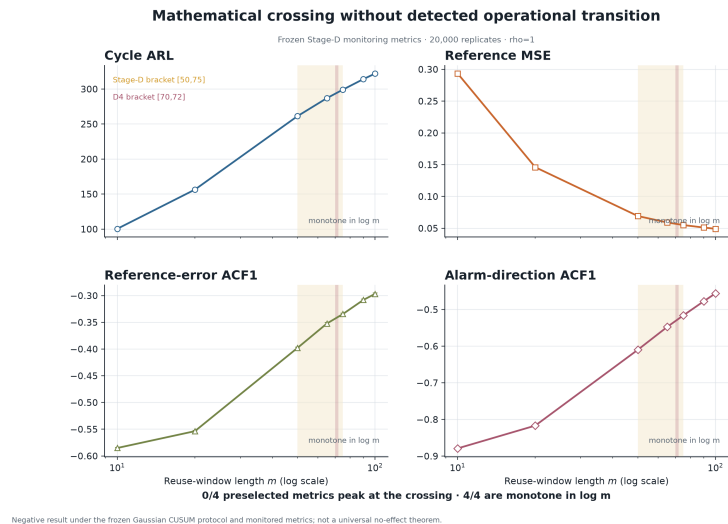


Figure 4. The four monitored operational metrics pass smoothly through the mathematical crossing.

The correct conclusion is narrow: no crossing-localized operational transition was detected for the frozen grid, shifts, and metrics. It is not a universal no-effect claim. Keeping this negative result visible prevents the mathematical boundary from being presented as stronger operational evidence than the study supports.

## 7. Historical campaign status

The Level-4 programme ran nine numbered campaigns. Their adjudicated verdicts are immutable and are not rewritten by any successor.

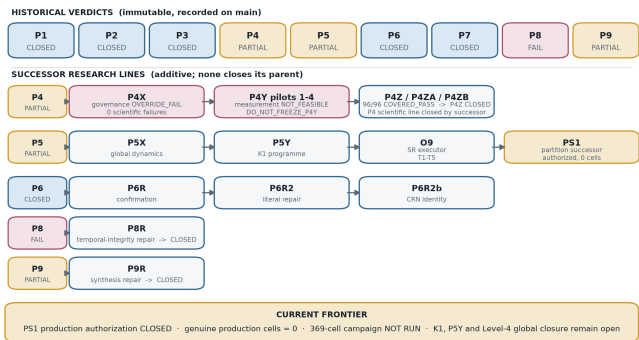
| Campaign   | Verdict | Scope of the recorded outcome                                                                       |
|------------|---------|-----------------------------------------------------------------------------------------------------|
| P1, P2, P3 | CLOSED  | Frozen derivative theorems, the SR derivative result, and the reuse-fraction stability map          |
| P4         | PARTIAL | Location-family theorem survives review; three preregistered numerical gates remain literally false |
| P5         | PARTIAL | Raw-mean identity and fixed-policy ergodicity survive; attraction and universality claims narrowed  |
| P6, P7     | CLOSED  | Safe-rebaselining scope and the operational-degradation consequence                                 |
| P8         | FAIL    | Cross-family window law rejected; G7 and the temporal-integrity gate fail                           |
| P9         | PARTIAL | Retrospective synthesis survives; several sub-claims conditional or unsupplied                      |
| P8R, P9R   | CLOSED  | Separate repair lineages behind pre-result anchors; P8 and P9 keep their own verdicts               |

## 8. Successor philosophy and the P4 line

A successor is a new campaign declared behind a pre-result temporal anchor that continues a research line. Supersession means later research continues the line; it never deletes an earlier campaign and never converts a PARTIAL or FAIL into a pass.

### Level-4 campaign chronology and successor lineage

A successor continues a research line; it never changes the earlier campaign's recorded verdict



The P4 successor line is published on the historical research branches p4x-feasibility-audit, p4y-prefreeze-pilot, p4y-pilot2-final, p4y-pilot3-heavy-stage1, p4y-pilot4-measurement, p4zb-skewnormal4-k7, p4z-rng-governance-repair and p4z-final-closure. P4Z CLOSED is a successor closure; P4 itself remains PARTIAL. Statuses are internal, scope-bound research designations.

Figure 5. Level-4 chronology and the successor lines that continue it.

The P4 line, published as historical research branches, illustrates why failed successors are retained. **P4X** reached 88/96 binding PASS with zero scientific failures, yet its verdict was overridden to FAIL on governance and sampling integrity: stage-2 top-up sharding executed four unauthorised blocks and 1,000,000 unauthorised paths, and precision attainment held only in expectation. **P4Y** ran four measurement-architecture pilots and concluded P4Y\_MEASUREMENT\_FEASIBILITY = NOT\_FEASIBLE with DO\_NOT\_FREEZE\_P4Y; no P4Y checkpoint was frozen and no P4Y production ran. The later **P4Z/P4ZA/P4ZB** line ran in three stages against the unchanged frozen gate, each recording its own self-verdict: P4Z P4Z\_NUMERICAL\_PASS\_AWAITING\_FORMAL\_OR\_GOVERNANCE (44/96 adjudicated), P4ZA P4ZA\_INCONCLUSIVE (92/96, four cells left open rather than widening the frozen K7 limit), and P4ZB P4ZB\_CLOSED (96/96 COVERED\_PASS). P4ZB\_CLOSED is the campaign's own self-verdict. Independent integrated adjudication of the line now records P4Z = CLOSED and P4\_SCIENTIFIC\_LINE = CLOSED\_BY\_LATER\_SUCCESSOR: all three of P4's originally failed gates are discharged, the 96-cell gate by the successor result and the outside-assumption and Gaussian-consistency gates by P4X obligations C4 and C5 admitted under an obligation-local Rule C that does not rehabilitate P4X as a campaign. Two disclosed defects remain on the record — the successor-added K7 precondition, redesigned twice with each version frozen before its own run, and a calibration/production RNG address overlap inside P4Z — both non-blocking because with the successor-added truncation term removed entirely all 96 cells still clear the frozen gate, at worst  $|z|$  1.7877 against 4.0 and worst relative discrepancy 0.0095 against 0.03. None of this closes P4 itself, which remains PARTIAL.

## 9. O9 certification architecture and governed negative results

The P5X/P5Y line carried the global-dynamics successor into the O9 SR certification executor, progressing through T1 candidate construction, T2 per-patch certification, T3 whole-cell aggregation, T4 curvature, and T5 one-cell and regional closure. The old cell-313 region of the frozen 316-cell SR cover exposed a genuine whole-cell curvature obstruction, and four separate routes to remove it were declared, executed, and falsified.

| Route                    | Recorded classification                     | Consequence for the design space                                    |
|--------------------------|---------------------------------------------|---------------------------------------------------------------------|
| Curvature successor      | CURVATURE_SUCCESSOR_NUMERICALLY_NOT_CLOSING | Predeclared kill condition failed at cell 313 for $m = 2, 3, 5$     |
| Operator-norm tightening | OPERATOR_NORM_TIGHTENING_INSUFFICIENT       | Tighter certified norms of the same operator did not close the cell |
| Aux3 third derivative    | AUX3_SR_FEASIBILITY_FAIL                    | The augmentation was not feasible under its own governance          |
| Endpoint route           | endpoint NOT_CLOSING, then strip micropilot | Endpoint gate later passed, but the frozen B_int line stayed open   |

These are governed scientific outcomes, not incidental engineering faults. Each removed a candidate mechanism from consideration, and together they are the reason the next successor changed the partition rather than the bound.

## 10. The PS1 partition successor

PS1 is a new additive K1 SR successor campaign. It keeps the theorem target, drift domain, m scope, per-cell B\_cover budget, precision, and degree unchanged, and declares one canonical 369-cell partition of the whole domain by a geometry-only rule ( $RHO\_CAP, r\_max = 1/25$ ) frozen before any successor result existed. It refines 25 parent cells and replaces the old-313 region with four deterministic successor cells. The union of successor cells is the original domain exactly.

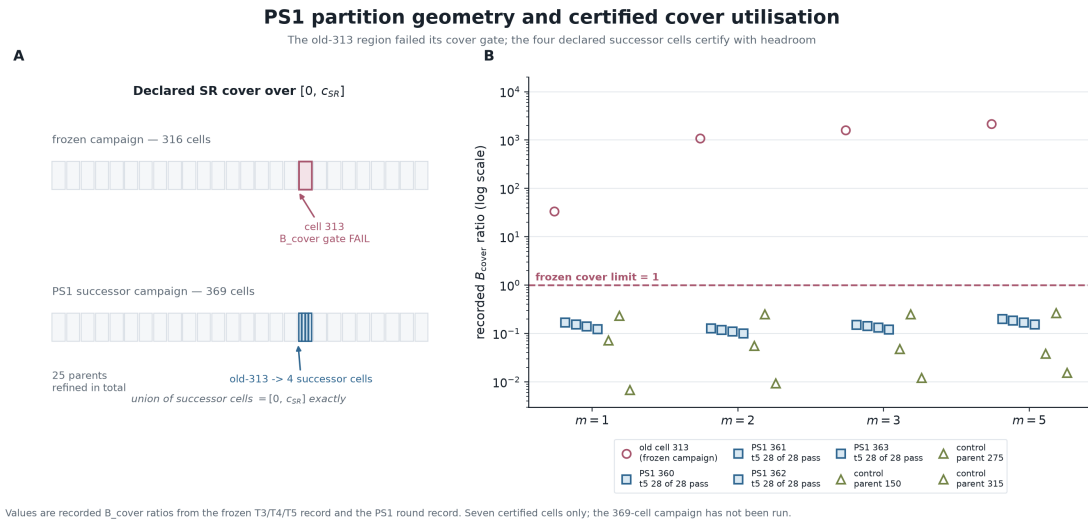


Figure 6. The failed old-313 cover ratios against the certified PS1 successor and control cells.

The four successor cells each recorded T5\_28\_OF\_28\_PASS, as did three certified control cells at parents 150, 275, and 315. Recorded  $B_{cover}$  ratios for the successor cells lie between 0.099 and 0.198 against a frozen limit of one, where the old cell 313 recorded ratios from 33.4 to 2174.5. The old 316-cell campaign and its cell-313 failure remain immutable historical evidence and are not claimed by PS1.

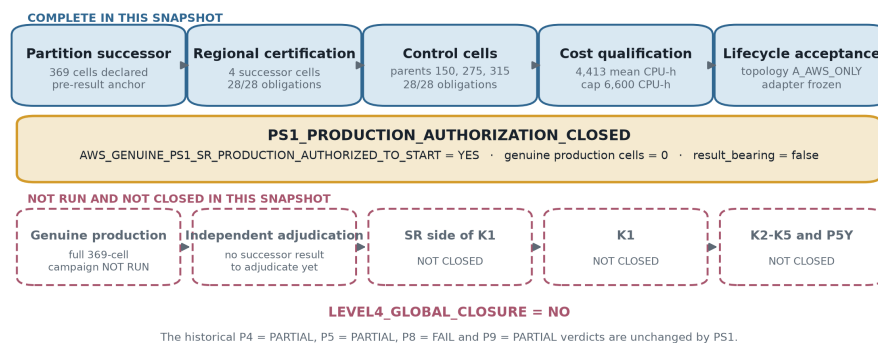
## 11. Cost qualification and production authorization

Requalification on the production unit measured a per-cell mean of 11.94 CPU-h across 13 real cells, giving a 369-cell projection of 4,413 CPU-h on mean groups and 4,642 CPU-h on the worst group, against a frozen global cap of 6,600 CPU-h. The executor reproduced 343/343 cell-patch records bit-identically to the committed certified records. Topology is A\_AWS\_ONLY; the Vultr host is NOT\_QUALIFIED\_FOR\_PS1. One pre-existing lifecycle-suite assertion fails because a historical torn-attempt ledger from 2026-09-10 remains in the namespace; it is byte-unchanged, records zero genuine cells, and is disposed as known\_pre\_existing\_only.

## 12. Exact current closure boundary

### Current PS1 boundary and remaining scientific steps

Authorization to start is a governance state, not a scientific result



Cost figures are the requalified projection recorded in PS1\_CONSTANTS.json; they are estimates, not measured campaign cost.

Figure 7. What PS1 has authorized, and what has not been run or closed.

The authoritative classification is PS1\_PRODUCTION\_AUTHORIZATION\_CLOSED, with AWS\_GENUINE\_PS1\_SR\_PRODUCTION\_AUTHORIZED\_TO\_START = YES. In this published snapshot the boundary is exact and narrow:

- genuine PS1 production cells = 0; production has **not started**;
- the full 369-cell genuine production campaign is **NOT RUN**;

- the SR side of K1 is **NOT CLOSED**; K1 is **NOT CLOSED**; P5Y is **NOT CLOSED**;
- LEVEL4\_GLOBAL\_CLOSURE = NO.

Authorization to start is a governance state, not a scientific result. The PS1 production namespace carries an empty result ledger at freeze and the recorded start command is intentionally not run.

### 13. Evidence and rigor summary

- **Human theorem:** analytic statements with explicit assumptions.
- **Lean-checked:** compiled formal proof spine for the encoded statements.
- **Arb-certified:** rigorous outward-rounded numerical enclosures for CUSUM and the authoritative SR detector.
- **Deterministic certificate:** interval result for the conditional-mean skeleton, not the noisy stochastic chain.
- **Empirical evidence:** frozen simulation and semi-real task evaluations.
- **Negative result:** a scoped pre-specified hypothesis not supported under its frozen design.
- **Governance and temporal integrity:** pre-result anchors, immutability checks, and audit records. This is process evidence, never a theorem.

### 14. Limitations

The stronger non-Gaussian extension remains partial. The work is not detector-general or distribution-general. Local deterministic results do not establish stochastic invariant behavior or an operational phase transition. Semi-real streams do not establish production readiness. The literature audit supports only a scoped N2 position, not absolute novelty or priority. PS1 evidence covers seven certified cells and says nothing about the remaining 362 cells of its declared universe, and its cost figures are projections rather than measured campaign cost. The Level-4 internal closure designation is a project checklist outcome, not an external academic standard.

### 15. Next scientific steps

Run the declared 369-cell PS1 campaign under its frozen cap, submit the result to independent adjudication, and only then consider the SR side of K1. K1 closure, the remaining K2-K5 work, and P5Y closure follow in that order. The P4 line needs no further work: its successor closure is adjudicated and its scientific line is closed by later successor evidence, with P4's own historical verdict left at PARTIAL. None of these steps is anticipated by the present snapshot.

### 16. Reproducibility and repository

The repository is internally closed under a pre-specified project checklist; that designation is not an external standard, publication decision, or peer-review result. Reproduce the frozen terminal snapshot with `bash level4/final_level4_closure/reproduce.sh` at tag `rebaseguard-level4-closed`. Reproduce the later SR certificate with `bash level4/closure_proofs/sr_derivative/certificate/reproduce_closed_upgrade.sh` at tag `rebaseguard-sr-gamma-certified`. Check the current presentation with `python3 scripts/verify_academic_presentation.py --no-diff-check` and `python3 docs/research_synthesis/verify_synthesis.py --no-diff-check`.

Repository: <https://github.com/flaggielover/ReBaseGuard> Research synthesis: `docs/research_synthesis/README.md`  
 Current successor status: `docs/research_synthesis/PS1_CURRENT_STATUS.md` Citation metadata: `CITATION.cff`  
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